Unit 3 Progress Check: FRQ

Include correctly labeled diagrams, if useful or required, in explaining your answers. A correctly labeled diagram must have all axes and curves clearly labeled and must show directional changes. If the question prompts you to "Calculate," you must show how you arrived at your final answer.

Quantity of Output	Total Cost
0	$12
1	$14
2	$18
3	$24
4	$32
5	$42
6	$54
7	$68

The table above shows the total cost function for a typical firm producing hats in a perfectly competitive market. The market price for hats is $9 per hat.

(a) Calculate the average variable cost of the fifth unit. Show your work.

(b) What is the firm's profit-maximizing quantity of hats? Explain using marginal analysis.

(c) If the rent of the building the firm occupies increases, what will happen to the firm's profit-maximizing quantity of hats in the short run? Explain.

(d) Draw a correctly labeled graph showing the firm's demand and marginal cost curves, and show the profit-maximizing quantity of hats, labeled Q*.

1. Respond to all parts of the question.

Include correctly labeled diagrams, if useful or required, in explaining your answers. A correctly labeled diagram must have all axes and curves clearly labeled and must show directional changes. If the question prompts you to "Calculate," you must show how you arrived at your final answer.

Assume that apples are produced in a perfectly competitive market. Grande's Orchard is a typical firm that grows and sells apples. Currently, Grande earns zero economic profit, and the market price of apples is $10 per bushel.

(a) Draw a correctly labeled graph showing Grande's demand curve, average total cost curve, and marginal cost curve, and show the profit-maximizing quantity, labeled Q_G .

(b) Suppose an increase in the popularity of apple cider increases the demand for apples. How will the increase in the demand for apples affect Grande's economic profit in the short run? Explain.

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(c) What will happen to Grande's economic profit in the long run? Explain.

2. Respond to all parts of the question.